

3. Geometric meaning

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Lecture due Feb 14, 2024 20:30 IST

Geometric meaning of real eigenvalues and eigenvectors

Recall that any $n \times n$ matrix $\mathbf{A}$ represents a function from $\mathbb{R}^n$ to $\mathbb{R}^n$. Therefore, an eigenvector $\mathbf{v}$ of $\mathbf{A}$, which satisfies the equation

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad \text{for some scalar } \lambda,$$

is a vector whose image under $\mathbf{A}$ is a scalar multiple of itself. When the eigenvalue λ is real, this means that an eigenvector is a vector $\mathbf{v}$ whose image lies on the line in $\mathbb{R}^n$ through $\mathbf{0}$ and $\mathbf{v}$, with the eigenvalue λ as the scaling factor.

Let us revisit the examples above.

Example 3.1 The function represented by $\begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ stretches every vector in $\mathbb{R}^3$ to **5** times its length but does not change its direction; hence, every vector is an eigenvector associated to the eigenvalue **5**.

Example 3.2 The function represented by the matrix $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

- stretches the eigenvector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ to **2** times (**2** is the corresponding eigenvalue) its length but does not change its direction;
- collapses the eigenvector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ to the **0** vector, since **0** is the corresponding eigenvalue;
- flips the eigenvector $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ across the origin to the other side of the **z**-axis without changing its length, since **-1** is the corresponding eigenvalue.

Example 3.3 The (function represented by the) matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

- does not change the direction or length of the eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ since the corresponding eigenvalue is **1**;
- flips the eigenvector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ across the origin to the other side of the line $y = -x$, since the eigenvalue is **-1**.

In each case, the image of the eigenvector lies on the same line as the eigenvector itself.

Matrix vector mathlet

The mathlet below shows the input and output vectors of a 2×2 matrix $\mathbf{A}$ and their relationship with the eigenlines. Recall from the course *Differential equations: 2 by 2 systems* that when the eigenvectors of an eigenvalue are all scalar multiples of just one eigenvector, then the line consisting of all the eigenvectors is called an **eigenline**.

To see the action of $\mathbf{A}$ on its eigenvector:

1. click on all three boxes "Show eigenlines," "Show eigenvalues," and "Show eigenvectors";

2. choose values of **A** (on the right) so that two eigenlines (green) are shown on the graph (on the left);
3. click on a point along the eigenlines (green on the graph) to select an eigenvector **v** (blue) as an input to **A**;
4. observe that the output **Av** (red) lies along the same line as **v**, with the corresponding eigenvalue, λ_1 or λ_2 , (bottom right) as scalar factors.

Observe that when the input vector **v** (blue) is **not** an eigenvector (i.e. not on an eigenline), the output **Av** (red) is not on the same line as **v**.

MATRIX VECTOR

1.00

10

5

0

-5

-10

y

-10

-5

0

5

10

x

1.00

1.00

Input vector

$v = \begin{bmatrix} 1.00 \\ 1.00 \end{bmatrix}$

2.00

1.00

Output vector

$Av = \begin{bmatrix} 2.00 \\ 1.00 \end{bmatrix}$

-6

-3

0

3

6

1.0

-6

-3

0

3

6

1.0

-6

-3

0

3

6

1.0

-6

-3

0

3

6

0.0

0

1

2

3

4

5

6

7

8

1.41

0

$\pi/2$

π

$3\pi/2$

2π

0.79

Show eigenlines

Show eigenvalues

Show eigenvectors

Projections

2/2 points (graded)

There are two eigenvalues for the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, namely 0 and 1.

Which of the following are eigenvectors associated to

the eigenvalue 1:

(Choose all that apply.)

☒

$\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$

☒

$\begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix}$

☐

$\begin{pmatrix} 0 \\ 0 \\ 1/2 \end{pmatrix}$

☒

$\begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix}$

the eigenvalue 0:

(Choose all that apply.)

☐

$\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$

☐

$\begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix}$

☒

$\begin{pmatrix} 0 \\ 0 \\ 1/2 \end{pmatrix}$

☐

$\begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix}$

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3/5

$\begin{pmatrix} -1/3 \\ 0 \end{pmatrix}$

Previous

$\begin{pmatrix} -1/3 \\ 0 \end{pmatrix}$

Next

☐

$\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

☐

$\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

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Note: Any vector of the form $\begin{pmatrix} a \\ b \\ 0 \end{pmatrix}$ is an eigenvector of $\mathbf{1}$. Geometrically, any vector lying on the xy -plane is unchanged by the matrix.

On the other hand, $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1/2 \end{pmatrix} = \mathbf{0}$, so $\begin{pmatrix} 0 \\ 0 \\ 1/2 \end{pmatrix}$ is an eigenvector of $\mathbf{0}$.

Note: $\begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix}$ for any number c is an eigenvector of $\mathbf{0}$. Geometrically, any vector lying on the z -axis is collapsed to the origin by the matrix.

Finally,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix},$$

which is not a scalar multiple of $\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$. So $\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ is not an eigenvector associated to any eigenvalue.

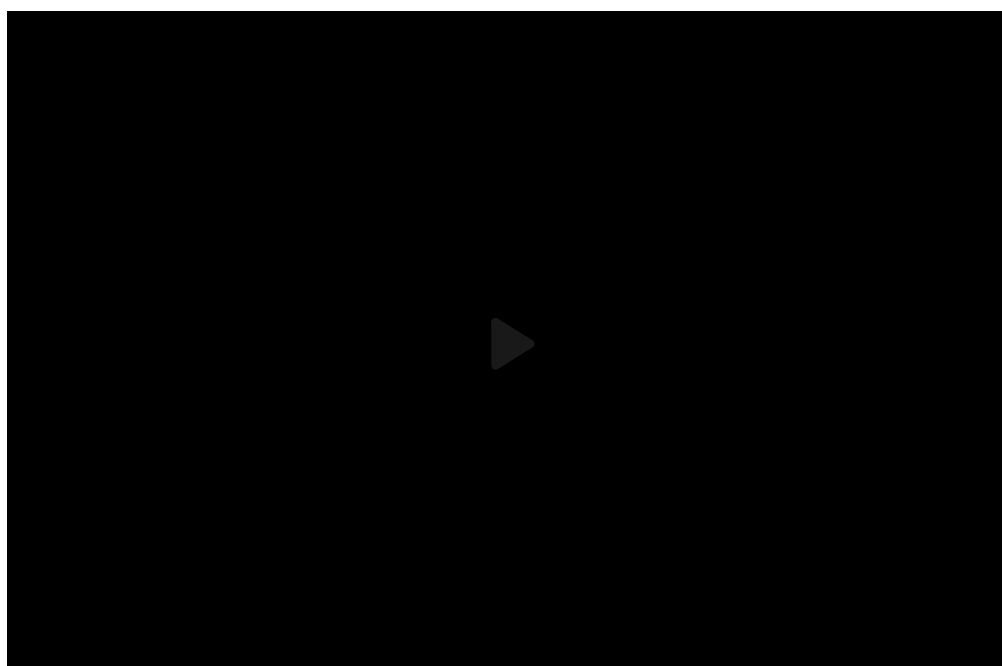
Therefore, the matrix represents the projection function that sends any vector in $\mathbb{R}^3$ to its “shadow” on the xy -plane.

Submit

You have used 1 of 3 attempts

i Answers are displayed within the problem

Introduction to eigenvectors



And I figure there's going to be a second eigenvalue

and eigenvector.

Now what about that?

What's a vector-- maybe we can just
like guess it--

a vector that the other--

actually, this one that I'm thinking
of

is going to be a vector that has eigenvalue minus 1.

That's going to be my other
eigenvalue for this matrix.

It's a nice positive or not negative matrix,

but an eigenvalue is going to come out negative.

 Calculator

▶

13:15 / 13:15

▶ 2.0x

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And can you guess--
spot the x that will work for that?
So I wanted a vector when I

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